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Executive Summary

This Work Integrated Learning (WIL) project examines the inefficiencies in the issuing of construction drawings at BBC Consulting Engineers, a mid-sized South African civil engineering consultancy based in the heart of Pietermaritzburg. The process is mission critical, affecting project delivery, client satisfaction, and organisational competitiveness.

The current (AS-IS) system relies on outdated CAD software, slow computers, unreliable printers, manual courier systems, and unstructured feedback loops. These inefficiencies result in frequent delays, versioning errors, and client dissatisfaction. Using business analysis methodologies including Business Genetics W5 modelling, root cause analysis, fishbone diagrams, gap analysis, Pareto analysis, and stakeholder engagement, the project identified systemic weaknesses in technology, process, human resources, and governance.

The key findings of the project reveal that technology obsolescence is the primary constraint, significantly limiting efficiency and reliability across the drawing issuance process. In addition, process gaps such as the absence of standardised templates, poor version control, and informal approval practices create systemic risks that undermine project delivery. Human resource limitations and insufficient training further burden engineers and administrative staff, while weak governance and chronic underinvestment in IT continue to prevent meaningful modernisation.

Based on these findings, the project recommends upgrading CAD infrastructure and computers and adopting cloud-based document management systems with version control. It further proposes standardising processes for requirements, approvals, revisions, and archiving to ensure consistency and control. Staff should be provided with training and supported through structured change management strategies to enable adoption of new systems and practices. Finally, governance must be strengthened by reframing IT as a strategic enabler rather than a cost centre and aligning IT investment with recognised governance frameworks such as COBIT and ISO.

A phased roadmap (short, medium, and long-term) is proposed, focusing on IT upgrades, workflow standardisation, and BIM enabled collaboration. Expected benefits include improved efficiency, reduced risk, higher client satisfaction, better staff morale, and enhanced competitiveness.

Ultimately, modernising the drawing issuance process will transform it from a liability into a strategic asset, ensuring BBC Consulting Engineers remains competitive in an evolving industry.

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Context Setting

Background of the Organisation

BBC Consulting Engineers is a mid-sized civil engineering consulting firm headquartered in Pietermaritzburg, KwaZulu-Natal. Established in the late 1990s, the organisation has grown steadily over the past two decades to become a respected player in the South African engineering consulting sector, with a particular reputation in hydrological, structural, and transportation engineering. The company employs approximately 50 staff, including 25 professionally registered engineers, 10 technicians, 5 administrative staff, and a support team of IT and clerical workers. BBC Consulting Engineers has an estimated annual turnover of R100 million, generated primarily through municipal infrastructure projects, private sector developments, and donor funded international projects (*Smith, 2023*).

The company is well known for its international engagements, particularly in hydrological projects such as dam upgrades and irrigation systems in Southern Africa. For example, BBC Consulting Engineers provided advisory services to Illovo Sugar in Malawi, upgrading irrigation networks to improve water efficiency. Such projects demand technical precision, timely delivery, and seamless communication with clients and contractors across borders. This positions the accuracy and timeliness of construction drawings as one of the most critical outputs of the firm.

Like many mid-tier consulting firms in South Africa, BBC Consulting Engineers faces challenges associated with technological adaptation, resource allocation, and the pressure to maintain competitiveness against larger international firms. Although the firm is technically strong, its reliance on outdated technology and traditional processes has created inefficiencies that limit its ability to deliver effectively. This is most evident in the process of issuing construction drawings to clients and contractors, which is the focus of this Work Integrated Learning (WIL) project.

The Issuing of Construction Drawings

At the core of BBC Consulting Engineers' operations is the preparation, review, approval, and issuing of construction drawings. These drawings form the contractual basis for implementation on site, providing detailed guidance for contractors and assurance to clients. The current process begins with the drafting of a drawing by a design engineer using AutoCAD 2010, one of the primary design tools still in use at the firm. Despite AutoCAD's robustness, the outdated computers and insufficient graphics capabilities at BBC result in frequent software crashes, lags, and corrupted files (*Jones & Lee, 2022*).

Once drafted, drawings are reviewed internally by a project manager, after which they are either approved or returned for revisions. Approved drawings are then printed using the office's Konica Minolta Bizhub printers. These machines, while once reliable, now produce frequent errors such as paper jams, misalignments, and inconsistent print quality. Engineers report that for every 10 large scale drawings attempted, at least three fail on the first attempt, requiring reprinting and consuming both paper and time.

The issuing process further involves administrative staff packaging printed drawings for courier to contractors or scanning and emailing them to clients. However, email limitations often make it difficult to send large files, while courier services introduce delays and additional costs. The lack of a centralised, cloud-based document management system further complicates matters, as version control becomes problematic. In several cases, contractors have worked from outdated drawing versions, leading to rework and disputes on site.

Overall, the process is fragmented, heavily reliant on outdated technology, and susceptible to error. This creates inefficiencies that ripple across projects, causing delays, budget overruns, and reputational risks.

Business Impact of Current Challenges

The consequences of inefficiencies in issuing construction drawings are significant. BBC Consulting Engineers has faced an increasing number of complaints from clients and contractors regarding late drawing submissions. Internal estimates suggest that

approximately 15% of client feedback in 2024 related to delays in receiving drawings, while around 10% of contractor disputes arose from versioning errors (Adams, 2024).

These inefficiencies have translated into measurable financial costs. For instance, in 2024, BBC Consulting Engineers reported project delays costing an estimated R1.5 million in unbillable staff hours, rework, and penalties imposed by clients for missed deadlines. Engineers frequently reported working overtime to redo lost work or correct errors caused by system crashes. Morale has been negatively impacted, with several staff members citing frustration and burnout in informal internal surveys.

From a strategic perspective, these inefficiencies pose a threat to the firm's long-term competitiveness. In the engineering consulting sector, timely delivery of accurate drawings is a basic expectation. Failure to consistently meet this expectation risks eroding client trust, damaging relationships with contractors, and losing bids to competitors who can demonstrate greater reliability and modern digital capabilities (*Brown, 2021*).

Beyond operational and financial risks, the inefficiencies expose BBC Consulting Engineers to legal and reputational risks. Outdated drawings issued to contractors can result in defective construction, contractual disputes, and potential litigation. In industries where liability risks are high, such failures can have lasting reputational damage that undermines decades of relationship building.

Value of Addressing the Issue

Addressing the inefficiencies in the issuing of construction drawings would create multiple forms of value for BBC Consulting Engineers. Firstly, it would improve operational efficiency by reducing rework, lowering overtime costs, and minimising errors. Studies have shown that engineering firms that digitise document management processes reduce turnaround times by up to 40% (*Taylor, 2020*).

Secondly, upgrading the process would enhance client satisfaction by ensuring timely delivery of accurate and standardised drawings. In an industry where repeat business and word-of-mouth referrals are critical, improved client trust could lead to significant business development opportunities (*Anderson, 2022*).

Thirdly, the adoption of modern systems and streamlined workflows would align BBC Consulting Engineers with industry best practices, improving its competitiveness in both local and international markets. Larger competitors often use cloud-based systems, BIM integration, and automated version control, failing to adopt similar processes risks leaving BBC behind (*Miller, 2023*).

Finally, process improvement would reduce risk exposure. Improved version control, standardisation, and secure digital systems would lower the likelihood of errors, disputes, and cyber vulnerabilities. With growing emphasis on sustainability and efficiency in engineering consulting, digitalisation would also reduce paper usage and associated costs, aligning the firm with broader environmental goals.

Business Model Canvas Application

The Business Model Canvas (BMC) provides a useful framework for situating the construction drawing issuing process within BBC Consulting Engineers' broader business model. To break down the canvas we incorporated all the business aspects into their respective categories. Please see Annexure B for a detailed visual demonstration of the business model canvas.

Value Proposition: The company's primary value lies in delivering accurate, high-quality engineering designs and drawings that enable clients and contractors to complete infrastructure projects on time and within budget. Delays and inaccuracies in drawing issuance undermine this value proposition (*Johnson, 2022*).

Key Activities: Core activities include drafting, reviewing, approving, printing, and issuing drawings. The inefficiencies in these activities reduce process efficiency and stakeholder satisfaction.

Key Resources: Current resources include AutoCAD 2010, outdated computers, and printers. These resources are not fit for purpose in supporting modern engineering requirements.

Key Partners: Clients (municipalities, developers), contractors, and IT suppliers. Poor communication and outdated tools weaken partnerships with clients and contractors.

Channels: Email, courier, and physical meetings. These channels are inefficient for large file transfers compared to modern cloud-based systems.

Customer Segments: Primary customers are municipalities, private developers, and contractors who expect professional service delivery. Dissatisfaction here directly threatens revenue.

Revenue Streams: Fee-for-service contracts based on project delivery. Delays in drawings reduce billable efficiency.

Cost Structure: High hidden costs due to rework, wasted time, and equipment maintenance. Investment in updated systems could reduce these costs long-term.

This analysis highlights that the drawing issuing process directly intersects with the value proposition, customer relationships, and cost structure of the business model, underscoring its strategic importance.

Porters Value Chain

Applying Porter's Value Chain framework further clarifies how the inefficiencies in issuing drawings affect organisational performance. Please see Annexure C for a detailed visual demonstration of Porters Value Chain.

Primary Activities:

Operations: The preparation and printing of drawings form part of operations. Inefficiencies here delay project outputs.

Outbound Logistics: The issuance of drawings to clients and contractors constitutes outbound logistics. Current reliance on outdated printers and courier services reduces efficiency.

Inbound Logistics: BBC Consulting Engineers depends on clients that need to design and construct something.

Marketing & Sales: Poor reputation for reliability undermines business development.

Service: Errors in issued drawings increase after-sales support requirements and disputes.

Support Activities:

Technology Development: Limited investment in IT has left the company vulnerable. Outdated systems hinder innovation.

Human Resource Management: Staff frustration and burnout due to inefficiencies impact morale and retention.

Firm Infrastructure: Weak governance of IT budgets and processes contributes to systemic inefficiency.

This analysis demonstrates that the drawing issuing process is not an isolated challenge but one that affects multiple parts of the value chain, from operations to service and reputation.

Problem Statement and Link to WIL Project

BBC Consulting Engineers is a technically competent organisation whose competitiveness is being undermined by inefficiencies in the issuing of construction drawings. The current process is heavily dependent on outdated computers, printing machines, and manual workflows. These limitations cause delays, errors, rework, client dissatisfaction, and reputational risks. The problem lies not only in technology but also in the absence of a streamlined, standardised, and digitised workflow for managing construction drawings.

This WIL project will focus on analysing the current state of the drawing issuance process, identifying root causes, and proposing sustainable process improvements. By applying business analysis models such as the W5 framework, root cause analysis, fishbone diagrams, gap analysis, Pareto prioritisation, and stakeholder analysis, the project will provide actionable recommendations. The aim is to improve efficiency, reduce risks, and align BBC Consulting Engineers with industry best practices, thereby securing its long-term competitiveness and value to stakeholders.

Theoretical Foundation

Introduction to Theoretical Foundation

The issuing of construction drawings within an engineering consulting firm is both a technical and critical process. Inadequacies in this process reflect broader challenges at the intersection of business process management, project management, and information systems. To address these inefficiencies in a structured and evidence based manner, it is important to ground the analysis in relevant academic and professional literature. This theoretical foundation provides a review of key theories and frameworks that underpin business process improvement in the context of BBC Consulting Engineers. The discussion integrates perspectives from the Business Analysis Body of Knowledge (BABOK), Business Process Management (BPM), Lean Six Sigma, IT-business alignment, and project management frameworks such as PMBOK and PRINCE2.

Business Analysis and Change Enablement

The International Institute of Business Analysis defines a business analyst as an “agent of change” responsible for identifying business needs and recommending solutions that deliver value to stakeholders (*IIBA, 2015*). Within this role, business analysts apply structured techniques for requirements elicitation, stakeholder engagement, and process modelling. In the case of BBC Consulting Engineers, the inefficiencies in issuing construction drawings represent a clear business need that requires intervention through business analysis practices.

BABOK highlights the importance of eliciting stakeholder perspectives through techniques such as interviews, workshops, and document analysis, which will form part of this project’s methodology (*Clark, 2021*). The application of business analysis also extends to defining both functional and non-functional requirements for improved processes, ensuring that any proposed solutions not only enable efficiency but also align with organisational strategy and stakeholder expectations (*Nguyen & Patel, 2020*).

Business Process Management (BPM)

BPM provides the conceptual basis for understanding and improving workflows such as the drawing issuing process. BPM emphasises the need to model, analyse, and redesign

business processes for better performance (*Harmon, 2019*). According to Dumas et al. (2021), BPM frameworks focus on aligning organisational processes with strategy, optimising resource use, and monitoring outcomes through key performance indicators.

The AS-IS and TO-BE models that will be developed in this WIL project are consistent with BPM principles, as they represent the current state and desired future state of the drawing issuing process. BPM also underscores the importance of continuous improvement and iterative evaluation, suggesting that improvements at BBC Consulting Engineers should not be treated as a once-off IT upgrade but as part of an ongoing cycle of evaluation, adjustment, and adaptation (*Smith, 2023*).

Lean Six Sigma and Waste Reduction

The Lean Six Sigma methodology designed to improve process efficiency by eliminating waste and reducing variability (*George, 2019*), was used in the context of BBC Consulting Engineers. It was used in conjunction with the Lean to emphasise the elimination of non-value-adding activities, while Six Sigma focuses on reducing defects and errors. Both perspectives are highly relevant to the challenges at BBC Consulting Engineers.

In the current drawing issuing process, waste is evident in the form of duplicated work (revisions due to lost files or outdated drawings), unnecessary waiting times (caused by computer crashes or courier delays), and over-processing (reprinting due to printer errors). Six Sigma's focus on defects also aligns with the issue of contractors working from incorrect versions of drawings, which represents a defect in service delivery. Studies indicate that firms applying Lean Six Sigma to documentation workflows have achieved improvements in turnaround time by up to 30% and reduced error rates significantly (*Johnson & Lee, 2022*).

Thus, Lean Six Sigma provides both a conceptual and practical basis for diagnosing inefficiencies and designing corrective measures in the drawing issuance process.

Information Systems and IT Business Alignment

Theoretical work on IT-business alignment highlights the strategic role of technology in enabling business value (*Henderson & Venkatraman, 1993*). Organisations that fail to align their IT capabilities with core business needs experience inefficiencies, wasted

resources, and competitive disadvantage. At BBC Consulting Engineers, the misalignment is clear, the outdated IT infrastructure is directly undermining the firm's ability to deliver its value proposition to clients.

Research by Luftman (2015) emphasises that IT should not be treated as a mere support function but as an enabler of strategic objectives. In consulting engineering, IT infrastructure is not peripheral but central, as it directly enables the production and dissemination of designs. Investment in updated CAD-compatible systems, digital document management, and cloud-based collaboration platforms is therefore not an optional upgrade but a strategic necessity (Williams, 2020).

Furthermore, theories of digital transformation stress the need for organisations to embrace technological change proactively. Firms that adopt digital document management systems report increased collaboration, improved version control, and reduced operational risks (Anderson, 2022). This literature underscores the imperative for BBC Consulting Engineers to transition from outdated tools to digital systems that support efficient and reliable drawing issuance.

Project Management Theories

The inefficiencies in issuing drawings can also be contextualised within project management theory. The PMBOK Guide defines project management as the application of knowledge, skills, tools, and techniques to project activities to meet requirements (PMI, 2021). One of the central frameworks within project management is the “triple constraint” of scope, time, and cost. Delays in drawing issuance directly threaten project timelines, increase costs through rework, and may require scope adjustments to recover lost time.

PRINCE2, another widely adopted project management methodology, highlights the importance of quality management, risk management, and stage boundaries (OGC, 2017). In this case, the issuing of drawings represents a critical stage boundary between design and implementation. Failures at this stage compromise the transition to construction, increasing project risks downstream.

The literature also highlights the growing emphasis on agile and hybrid project management approaches in engineering contexts (Miller, 2021). Agile emphasises

iterative delivery, stakeholder collaboration, and responsiveness to change. While full agile adoption may be impractical in traditional engineering, incorporating agile-inspired practices such as incremental drawing reviews or collaborative digital platforms could enhance responsiveness at BBC Consulting Engineers.

Integration of Theories BBC Consulting Engineers' Context

The theoretical perspectives reviewed above converge to provide a strong foundation for this WIL project. Business analysis theory frames the role of the analyst as a change enabler who identifies needs and proposes solutions. BPM and Lean Six Sigma provide structured approaches for diagnosing inefficiencies and designing process improvements. IT business alignment theory underscores the strategic importance of investing in modern systems. Project management frameworks highlight the implications of drawing issuance inefficiencies for scope, time, and cost.

Together, these theories create a robust framework for investigating and addressing the inefficiencies at BBC Consulting Engineers. They suggest that improvements should not focus solely on technology but must also consider processes, people, and strategy. This holistic theoretical foundation will guide the subsequent analysis of the AS-IS process, the application of W5 models, and the development of targeted recommendations for improvement.

Business Process Models (AS-IS)

Introduction to W5 Models

The W5 framework provides a structured way to capture and analyse the current state business process of issuing construction drawings at BBC Consulting Engineers. By examining the process through the lenses of WHAT, WHO, WHICH, WHERE, and HOW, inefficiencies become visible and opportunities for improvement can later be explored.

WHAT Model

The WHAT Model was developed using the Business Genetics W5 software to get the sense of the process of issuing drawings to stakeholders. The activities were detailed in accordance with the WHAT model rules, using the verb and accompanying noun. Please see Annexure D for a detailed visual demonstration of the WHAT Model.

Level 1 Main Activity: Issuing of Drawings

Level 2 Activities:

- Engage Client - Hold scope, understand drawing needs.
- Gather Requirements - Collect design criteria, investigate required standard.
- Draft Preliminary Drawing - Select software, create initial sketch, check prelim, attend to changes.
- Submit Preliminary Drawing - Convert drawings to PDF, send prelim to client.
- Revise Client Feedback - Record comments, revise drawings, conduct final CAD checks, validate changes.
- Obtain Client Approval - Secure sign-off for construction use.
- Draft Final Drawing - Produce final design, compile submission folder, PDF drawings, print drawings.
- Prepare Distribution Copy - Generate print-ready PDFs, assemble physical sets.
- Send Distribution Copies - Email PDFs, dispatch physical sets, confirm receipt.

WHO Model

The WHO Model was populated to outline the key stakeholders within the organisation based on each activity that is conducted within the drawing issuance process. Each step involves dependencies, and delays in one role often cascade into the rest of the process. Please see Annexure E for a detailed visual demonstration of the WHAT Model.

- Engage Client: Design Engineer, Project Manager, Client Representative.
- Gather Requirements: Design Engineer, Client Representative, Technician (supporting technical inputs).
- Draft Preliminary Drawing: Design Engineer, Technician.

- Submit Preliminary Drawing: Design Engineer, Administrative Staff (assist with formatting).
- Revise Client Feedback: Design Engineer, Project Manager, Client Representative.
- Obtain Client Approval: Project Manager, Client Representative.
- Draft Final Drawing: Design Engineer, Technician, Administrative Staff.
- Prepare Distribution Copy: Administrative Staff, IT Support (assist with large files).
- Send Distribution Copies: Administrative Staff, Courier Service, Client Representative, Contractor Representative.

This WHO model shows the collaboration across technical, administrative, and external stakeholders.

WHICH Model

The tools, systems, and documents used in each step:

- Engage Client: MS Teams/Zoom, meeting minutes, notebooks.
- Gather Requirements: Project briefs, design codes, client specifications, spreadsheets.
- Draft Preliminary Drawing: AutoCAD 2010, desktop computers, shared network drive.
- Submit Preliminary Drawing: Microsoft Outlook, PDF conversion software.
- Revise Client Feedback: CAD software, marked-up PDFs, tracked emails.
- Obtain Client Approval: Email correspondence signed approval forms, PDF files.
- Draft Final Drawing: AutoCAD, plotter printer, submission checklists.
- Prepare Distribution Copy: PDF tools, printers, scanners, binding machines.
- Send Distribution Copies: Email, courier services, courier tracking logs.

The WHICH model shows reliance on outdated IT systems and hardware, increasing risks of error, duplication, and file corruption. Please see Annexure F for a detailed visual demonstration of the WHICH Model.

WHERE Model

Physical and digital locations for each activity:

- Engage Client: Office boardroom, client office, or virtual meeting platform.
- Gather Requirements: Office desk, client's site, digital repositories.
- Draft Preliminary Drawing: Design office workstation.
- Submit Preliminary Drawing: Shared drive, project manager's office, email system.
- Revise Client Feedback: Design office, client's office, digital platforms.
- Obtain Client Approval: Client office, email exchange.
- Draft Final Drawing: Design office workstation, printing room.
- Prepare Distribution Copy: Printing room, admin desk.
- Send Distribution Copies: Office dispatch area, courier service, client site, contractor site.

The WHERE model shows a fragmented mix of physical and digital touchpoints, with inefficiencies in file transfer and physical delivery. Please see Annexure G for a detailed visual demonstration of the WHAT Model.

HOW Model

The HOW model integrates all elements into a process flow:

The process begins with Engage Client, where the design engineer and project manager confirm scope and drawing needs (WHO: Engineer, Manager, Client; WHICH: MS Teams, notes; WHERE: Office/virtual). Requirements are then collected during Gather Requirements using client specifications, design codes, and project briefs.

Next, the design engineer and technician Draft Preliminary Drawing in AutoCAD 2010. Once complete, the drawing is converted to PDF and Submit Preliminary Drawing is sent

to the client and project manager via email. Clients review and provide comments, triggering Revise Client Feedback, where engineers incorporate revisions and validate final CAD checks.

When acceptable, the project manager requests the client to Obtain Client Approval, securing sign-off for construction use. Engineers then Draft Final Drawing, compiling submission folders, generating PDFs, and printing sets. Admin staff then Prepare Distribution Copy, creating print-ready PDFs and assembling physical sets.

Finally, admin staff Send Distribution Copies, emailing PDFs, dispatching physical sets via courier, and confirming receipt with clients and contractors. The process ends only once acknowledgements are received.

This HOW narrative highlights multiple inefficiencies: manual printing and scanning, email bottlenecks for large files, courier delays, and lack of a centralised document management system. These pain points increase costs, delays, and risks of versioning errors. Please see Annexure H for a detailed visual demonstration of the HOW Model.

Conclusion on AS-IS Models

The revised W5 analysis reveals that while BBC Consulting Engineers' drawing issuance process covers all necessary technical and administrative steps, it remains highly manual, fragmented, and dependent on outdated tools. The WHAT model demonstrates a long chain of interdependent activities with numerous handoffs. The WHO model illustrates reliance on multiple stakeholders, often causing delays when one party is unavailable. The WHICH and WHERE models reveal outdated systems and inefficient reliance on physical spaces. The HOW narrative ties these issues together, showing a process vulnerable to delays, duplication, and client dissatisfaction.

Business Process Analysis Techniques

Issues List

The issues identified in the issuing of drawings process at BBC Consulting Engineers emerge directly from the WHAT model activities. They represent recurring pain points experienced by engineers, administrative staff, project managers, clients, and contractors. These issues are summarised below:

Client Engagement Issues

Client engagement issues can arise from several factors. Meetings often lack structured agendas, which leads to incomplete requirement capture. Additionally, misunderstandings about the project scope can occur when client expectations are not formally documented. Furthermore, the absence of a central communication log increases the risk of conflicting instructions, complicating the engagement process.

Requirement Gathering Issues

Design criteria are sometimes vague or inconsistently provided by clients, leading to confusion in the project. Furthermore, the lack of standardized requirement templates results in omissions in technical specifications, which can hinder the development process. Additionally, investigating codes and standards becomes time-consuming due to the absence of a central library, complicating compliance and design efforts.

Preliminary Drawing Issues

Preliminary drawing issues can significantly impact productivity and efficiency. Outdated AutoCAD software frequently crashes, leading to lost work and frustration. Additionally, slow desktop computers further reduce productivity during the initial drafting phase. Engineers also find themselves spending excessive time checking for errors, often without the benefit of peer support, which can delay project timelines and affect overall quality.

Submission of Preliminary Drawings Issues

Feedback and revision issues can create significant challenges in the project workflow. Client feedback is often unstructured, manifesting as handwritten notes or scattered emails, which complicates the revision process. Additionally, the absence of a version control system leads to confusion with multiple revisions, making it difficult to track changes accurately. Consequently, engineers spend a considerable amount of time validating these revisions, which ultimately reduces overall efficiency and productivity.

Approval Issues

Approval issues can significantly impede project progress. Delays in obtaining client approval often arise from scheduling conflicts, which can prolong timelines. Furthermore, the lack of a clear approval workflow creates uncertainty regarding responsibility, leaving team members unsure of who is accountable for securing approvals. Additionally, sign-offs are frequently informal, relying on emails rather than standardized authorization forms, which can lead to misunderstandings and complications in the approval process.

Final Drawing Issues

Final drawing issues can lead to significant inefficiencies in project completion. Printing errors, such as misalignments and paper jams, often result in repeated reprints, wasting both time and resources. Additionally, submission folders frequently lack consistency in content and formatting, making it difficult to organize and retrieve documents. Moreover, drawings are sometimes lost in shared drive folders due to poor naming conventions, further complicating access and increasing the risk of errors in final submissions.

Distribution Issues

Distribution issues can create significant obstacles in ensuring timely project execution. Courier delays often lead to late delivery of physical sets to contractors, which can disrupt schedules. Additionally, email dispatches are frequently hindered by large file sizes, making it challenging to send documents efficiently. The lack of delivery tracking further complicates matters, as it becomes difficult to confirm whether recipients have received materials in a timely manner, increasing the risk of miscommunication and delays.

Archival Issues

Archival issues can significantly hinder efficient project management. Digital archives are often disorganized, containing duplicate and outdated files that complicate access to current information. Meanwhile, physical archives consume valuable space and are difficult to search, making it challenging to locate necessary documents. As a result, retrieval delays frequently occur when older drawings are requested for reference, slowing down the workflow and increasing the risk of errors in ongoing projects.

These issues illustrate inefficiencies across every phase of the process, from client engagement to archival, underscoring the systemic nature of the problem.

Root Cause Analysis

Root cause analysis provides a structured approach to diagnosing the underlying reasons for these inefficiencies. The identified issues suggest four major categories of root causes:

Technological Obsolescence

Technological obsolescence poses significant challenges for project efficiency and effectiveness. The use of an outdated version of AutoCAD, along with computers that are ill-suited for modern projects, limits design capabilities and slows down workflows. Additionally, legacy printers and scanners often contribute to frequent breakdowns, resulting in low productivity and frustration among team members. Furthermore, the absence of digital collaboration platforms increases reliance on manual processes, making communication and teamwork more cumbersome and less efficient.

Process Gaps

Process gaps can significantly hinder operational efficiency and project success. The absence of standardized templates for requirement gathering, drawing submissions, and approvals creates inconsistencies that complicate workflows. Additionally, the lack of a formal workflow for client approvals often results in delays, prolonging project timelines. Furthermore, poor version control contributes to confusion among team members, leading to rework and inefficiencies that can derail progress and increase costs.

Human Resource Limitations

Human resource limitations can significantly impact project efficiency and overall productivity. Administrative staff often lack adequate training in document handling and the use of digital tools, which hampers their ability to support the team effectively. This situation results in engineers being burdened with administrative tasks, such as validating PDF conversions, diverting their focus from core engineering work. Additionally, project managers are stretched thin, which slows approval turnaround times and can lead to delays in project progress. These limitations hinder the organization's ability to operate smoothly and meet deadlines.

Organisational Culture and Governance

Organisational culture and governance can present substantial barriers to progress and efficiency. There is often resistance to adopting new technologies, primarily due to entrenched work practices that prioritize tradition over innovation. Additionally, limited IT governance and underinvestment in digital transformation hinder the organization's ability to modernize and streamline operations. The prevailing informal communication culture further complicates matters, resulting in a reliance on emails and ad hoc approvals, which can lead to misunderstandings and delays in decision-making. These factors collectively impede the organization's ability to adapt and thrive in a rapidly evolving environment.

These categories form the foundation for targeted improvement strategies, as they highlight systemic inefficiencies beyond isolated technical problems (*Smith, 2023*).

Fishbone Diagram

A fishbone (Ishikawa) diagram was developed to classify potential causes of inefficiencies under six broad categories:

- Methods: Non-standardised workflows, reliance on manual printing, lack of documented approval processes.
- Machines: Outdated computers, AutoCAD 2010, failing printers and scanners.
- People: Insufficient staff training, resistance to change, overburdened project managers.

- Materials: Inconsistent submission folders, disorganised archives, poor-quality printouts.
- Environment: Physical distance between offices and clients, reliance on courier services.
- Management: Limited investment in IT, weak governance of document processes.

The diagram (Annexure A) visually illustrates how inefficiencies arise not from a single source but from interconnected categories. Please see Annexure I for a detailed visual demonstration of the Fishbone Diagram.

Gap Analysis

Gap analysis compares the AS-IS state with the desired TO-BE state:

- AS-IS: Manual workflows, outdated IT infrastructure, courier-dependent delivery, poor version control.
- TO-BE: Digital workflows, updated CAD and hardware, cloud-based collaboration, automated version control.

Key Gaps Identified:

- Lack of a centralised digital document management system.
- Insufficient staff training in modern software and collaboration tools.
- Absence of structured approval workflows.
- Poor client engagement documentation.

Closing these gaps would align BBC Consulting Engineers with industry best practices and improve efficiency, accuracy, and client satisfaction (*Taylor, 2020*).

Why Analysis

The “5 Whys” technique was applied to the recurring issue of late delivery of drawings:

- Why 1: Drawings are delayed because courier services are slow.
- Why 2: Courier is required because files are too large for email.

- Why 3: Files are too large because outdated software cannot compress them efficiently.
- Why 4: Outdated software is still used because IT budgets have not been prioritised.
- Why 5: Budgets are neglected because management underestimates the strategic value of IT.

This analysis shows that the ultimate root cause of delays lies not in courier inefficiency but in **management's underinvestment in IT infrastructure**.

Pareto Analysis

A Pareto analysis was conducted to prioritise issues by frequency and impact. Based on staff interviews and client complaints, the following distribution emerged:

- Outdated computers and software: 30% of issues.
- Printer and scanner errors: 20% of issues.
- Version control failures: 20% of issues.
- Client feedback delays: 15% of issues.
- Courier delivery delays: 10% of issues.
- Archival inefficiencies: 5% of issues.

The Pareto chart (Annexure B) demonstrates that addressing the top three issues (computers/software, printers/scanners, version control) would eliminate approximately 70% of process inefficiencies. Please see Annexure J for a detailed visual demonstration of the Pareto Analysis.

Mock Interviews and Workshops

Mock Interviews

Structured interviews were conducted with representative staff to elicit insights:

- Interview with Design Engineer: Reported frequent AutoCAD crashes and wasted hours redoing lost work. Expressed frustration with manual checks and absence of peer review systems.
- Interview with Administrative Officer: Highlighted difficulties in handling large files, frequent printer malfunctions, and lack of training in document management.
- Interview with Project Manager: Cited delays in approvals due to excessive workload and absence of streamlined workflows. Stressed the need for a digital approval system.
- Interview with Client Representative: Expressed dissatisfaction with delayed drawings and confusion over multiple versions. Requested standardised digital submission formats.

Findings confirm that inefficiencies affect all stakeholders, with consistent calls for digital solutions and standardised workflows. Please see Annexure K for a detailed template used for the Mock Interviews conducted.

Mock Interviews

A workshop was convened with engineers, admin staff, and project managers. The objectives were to discuss pain points and brainstorm potential improvements.

Workshop Highlights:

- Engineers requested upgraded computers and cloud-based CAD platforms.
- Admin staff suggested training on document management systems.
- Project managers advocated for a structured approval workflow with digital sign-offs.
- Consensus reached that courier reliance should be reduced by adopting cloud-based sharing systems.

Minutes of this workshop (Annexure C2) reflect both the frustrations of staff and their readiness to adopt modern solutions if provided with training and support.

Stakeholder Context

Stakeholder analysis reveals the relative influence and interests of different groups:

- High Power, High Interest: Clients and contractors — demand timely, accurate drawings.
- High Power, Low Interest: Senior management — controls budgets but underestimates IT importance.
- Low Power, High Interest: Engineers and admin staff — directly affected by inefficiencies but lack decision-making authority.
- Low Power, Low Interest: IT support staff — have limited influence but play operational roles.

A power/interest grid (Annexure D) demonstrates that success depends on engaging management to prioritise IT investment; while ensuring clients remain satisfied and staff are equipped to adopt new systems.

Conclusion on Business Process Analysis

The combined application of issues listing, root cause analysis, fishbone diagrams, gap analysis, why analysis, Pareto prioritisation, interviews, workshops, and stakeholder analysis reveals that the inefficiencies in the issuing of drawings at BBC Consulting Engineers are systemic. They stem from a combination of outdated technology, weak processes, insufficient training, and limited management prioritisation of IT.

While individual issues such as courier delays or printer jams appear problematic on the surface, deeper analysis demonstrates that technological obsolescence and governance weaknesses are the ultimate root causes. Addressing these areas will yield the greatest impact on efficiency, client satisfaction, and competitiveness.

Results, Conclusions, and Recommendations

Introduction

The preceding sections have provided an extensive exploration of the inefficiencies associated with the issuing of construction drawings at BBC Consulting Engineers. Through the application of business process modelling (W5), issues analysis, root cause

analysis, gap analysis, fishbone diagramming, Pareto prioritisation, and stakeholder engagement techniques, a comprehensive picture of the AS-IS state has emerged. The findings highlight that inefficiencies stem not from isolated technical flaws but from systemic weaknesses in technology, process, governance, and culture. This section synthesises these results, articulates key conclusions, and provides structured recommendations to guide the organisation toward a more efficient and competitive TO-BE state.

Results of the Analysis

Key Issues Identified

The analysis revealed that inefficiencies were present across all stages of the drawing issuance process. Client engagement and requirement gathering were undermined by unstructured communication practices and the absence of standardised templates, which often resulted in misaligned expectations and incomplete specifications. Preliminary drawing preparation was further slowed by outdated AutoCAD software and underpowered computers, creating frequent delays and rework.

The submission and revision stages were similarly constrained by email size limitations, poor version control, and unstructured client feedback, which collectively disrupted workflow continuity and increased the likelihood of errors. Approval processes lacked formalised workflows, leading to prolonged delays and inconsistent sign-offs across projects. Furthermore, the production and distribution of final drawings were hampered by unreliable printers, courier delays, and ineffective file management systems.

Finally, archival processes were disorganised and lacked effective digital search or retrieval functionality, making it difficult to locate and track previous versions of drawings. Collectively, these inefficiencies highlighted systemic weaknesses that compromised both operational performance and client satisfaction.

Root Causes

The analysis indicated that the root causes of the inefficiencies in the drawing issuance process could be categorised into four main areas. First, technological obsolescence was

a critical factor, as the organisation continued to rely on outdated CAD systems, underpowered computers, unreliable printers, and inefficient channels such as email and courier services. These limitations significantly constrained productivity and undermined the quality and timeliness of outputs.

Second, process gaps were evident in the absence of standardised templates, formal approval workflows, and version control mechanisms, all of which led to inconsistent practices and frequent errors. Human resource limitations compounded these issues, with administrative staff lacking sufficient training, project managers being overburdened, and engineers being required to undertake administrative tasks outside their core responsibilities. Finally, organisational governance and culture were identified as structural barriers, marked by weak IT governance, limited financial investment in technology, and a resistance to adopting new systems and processes. Collectively, these factors reinforced systemic inefficiencies and hindered sustainable process improvement.

Gap Analysis Results

The gap between the AS-IS and TO-BE state was particularly evident in four areas:

- Lack of digital document management systems (AS-IS) vs. need for cloud-based platforms (TO-BE).
- Manual approvals and ad hoc communication (AS-IS) vs. structured, standardised workflows (TO-BE).
- Outdated infrastructure (AS-IS) vs. modern CAD-compatible hardware and software (TO-BE).
- Fragmented client engagement (AS-IS) vs. formalised communication protocols (TO-BE).

Pareto Prioritisation

Pareto analysis showed that addressing just three issues, outdated IT systems, printer/scanner failures, and version control weaknesses would resolve approximately 70% of inefficiencies. This prioritisation provides a practical focus for resource allocation.

Stakeholder Insights

Interviews and workshops demonstrated broad stakeholder dissatisfaction with the current process but also revealed willingness to adopt improvements. Engineers, project managers, and administrative staff unanimously identified digitalisation as the most urgent need. Clients expressed frustration at delays and versioning errors but indicated strong support for modernised digital submissions. Senior management, though less engaged, acknowledged that competitive pressures required IT investment.

Conclusions

The analysis produced several significant conclusions regarding the issuing of drawings process at BBC Consulting Engineers. Firstly, it became clear that the issuing of drawings is mission-critical; it is not simply an administrative task but a process that underpins project delivery, client satisfaction, and the overall reputation of the firm. Inefficiencies in this area compromise the entire value chain and directly impact business performance.

Secondly, technology emerged as the central constraint, with outdated hardware and software identified as the most critical root causes. Without substantial technological upgrades, other process reforms would have only limited effectiveness. Thirdly, the study highlighted the lack of process standardisation and control. The absence of structured workflows and version control mechanisms has created systemic risks, such as client disputes and costly rework.

Fourthly, stakeholder analysis demonstrated that most staff and clients are ready for change and willing to adopt digital solutions, provided adequate training and support are offered, although some resistance remains. Finally, the findings underscored that management must reframe IT as a strategic enabler rather than a cost centre. Underinvestment in IT has stemmed from a narrow financial perspective, but repositioning technology as essential to competitiveness is crucial for the firm's long-term success. In conclusion, addressing the inefficiencies in issuing drawings requires a holistic approach that combines technological upgrades, process redesign, targeted training, and cultural change.

Recommendations

Technological Recommendations

A primary recommendation is to upgrade the organisation's CAD infrastructure. This should involve adopting the latest version of AutoCAD or an equivalent Building Information Modelling (BIM) platform, supported by high-performance computers with enhanced graphics capabilities. Ensuring that all software licences are up to date and properly supported will further improve reliability and productivity.

The implementation of a digital document management system (DDMS) is equally critical. Cloud-based platforms such as Autodesk BIM 360 or SharePoint should be adopted to enable secure, centralised storage. These systems would also provide built-in version control, structured access management, and permissions-based access for both clients and contractors.

Modernising printing and scanning systems is also essential to reduce operational inefficiencies. Outdated Konica Minolta printers should be replaced with high-capacity plotters capable of handling large-format drawings. Additionally, the introduction of digital signing and stamping tools will reduce reliance on physical copies and streamline the approval process.

Finally, communication channels should be upgraded to support more effective collaboration. The organisation should move away from reliance on email and instead adopt secure file-sharing platforms. Integrating automated notifications for submissions and approvals will ensure timely communication and reduce the risk of delays.

Process Recommendations

A key process recommendation is to standardise requirement gathering. This can be achieved by developing structured templates for capturing client requirements and introducing comprehensive checklists for design codes and specifications. Such measures will ensure that the information collected at the start of projects is both complete and consistent.

Another priority is to formalise approval workflows. The introduction of structured digital approval processes, supported by electronic signatures, will provide transparency and

accountability. Establishing clear turnaround times for client responses will also help reduce bottlenecks and delays.

Feedback management should likewise be streamlined. The use of standardised feedback forms will make it easier to capture and interpret client revisions. Incorporating revision history directly within the digital document management system (DDMS) will improve traceability and minimise errors caused by multiple uncontrolled versions.

Finally, archival systems must be improved to support efficiency and retrieval. Legacy archives should be digitised and migrated to cloud-based storage. In addition, consistent naming conventions and metadata standards must be applied to ensure that drawings can be quickly and accurately located when needed

Human Resource Recommendations

Human resource improvements should focus on training, role clarity, and change management. Administrative staff must be trained in the use of digital document management systems (DDMS), PDF handling, and digital collaboration, while engineers should receive ongoing CAD upskilling and project managers should be trained in digital workflow management. Clearer role definition is also essential, reducing the administrative burden on engineers by empowering trained support staff and freeing project managers from bottlenecks through automated approvals. To support adoption, structured change management is required, including training sessions, pilot projects, and phased rollouts. Celebrating short-term wins, as suggested by Kotter's change framework, will further build momentum and reduce resistance to change (Kotter, 2012).

Organisational and Governance Recommendations

Governance improvements should focus on repositioning IT as a strategic asset, ensuring that technology investment is fully integrated into the organisation's strategic plan. This requires appointing an IT champion or establishing a steering committee to oversee digital transformation initiatives. To strengthen accountability, IT governance frameworks such as COBIT or ISO 38500 should be adopted, with performance monitored and reported alongside financial and project metrics. In addition, client communication protocols must be formalised through the creation of central logs of client interactions and the

establishment of service-level agreements (SLAs) to ensure that drawing issuance timelines are consistently met.

Implementation Roadmap

To ensure manageable change, the recommendations should be implemented in a phased manner. In the short term (0–6 months), the focus should be on procuring upgraded CAD software and high-performance computers, providing staff training on digital tools, and standardising requirement and feedback templates to create a more structured process foundation. In the medium term (6–18 months), the organisation should prioritise implementing a digital document management system (DDMS) with version control, digitising legacy archives to improve accessibility, and formalising digital approval workflows to enhance efficiency and accountability.

In the long term (18–36 months), BBC Consulting Engineers should transition fully to BIM-enabled digital collaboration with clients and contractors, embed IT governance frameworks such as COBIT or ISO 38500 to strengthen oversight, and adopt continuous improvement practices by applying Lean Six Sigma principles to monitor and refine processes on an ongoing basis.

Expected Benefits

The proposed recommendations are expected to generate significant benefits across multiple dimensions of the organisation. From an efficiency perspective, the improvements will reduce delays, enable faster turnaround times, and lower overtime costs. Enhanced reliability and standardisation will also strengthen client satisfaction by building greater trust and consistency in deliverables. Risk reduction is another key outcome, as stronger version control and structured approvals will minimise disputes and costly rework.

Employee morale is expected to improve as frustrations decrease, leading to higher job satisfaction and stronger staff retention. Furthermore, modernised workflows will provide BBC Consulting Engineers with a competitive advantage by positioning the firm as a credible rival to larger consultants. Finally, the adoption of digital processes and energy-efficient systems will reduce reliance on paper and align operations with sustainability objectives.

Conclusion

This WIL project has shown that inefficiencies in issuing construction drawings are systemic and multifaceted. By combining business process modelling, analytical tools, and stakeholder input, it is evident that the root causes lie in outdated technology, weak processes, insufficient training, and limited IT governance.

The recommendations provided are holistic, addressing not only technology but also process, people, and governance dimensions. Implementing these measures will enable BBC Consulting Engineers to transform the drawing issuance process from a liability into a strategic strength. Such improvements will enhance efficiency, reduce risk, increase client satisfaction, and secure the firm's long-term competitiveness in the engineering consulting industry.

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Annexures

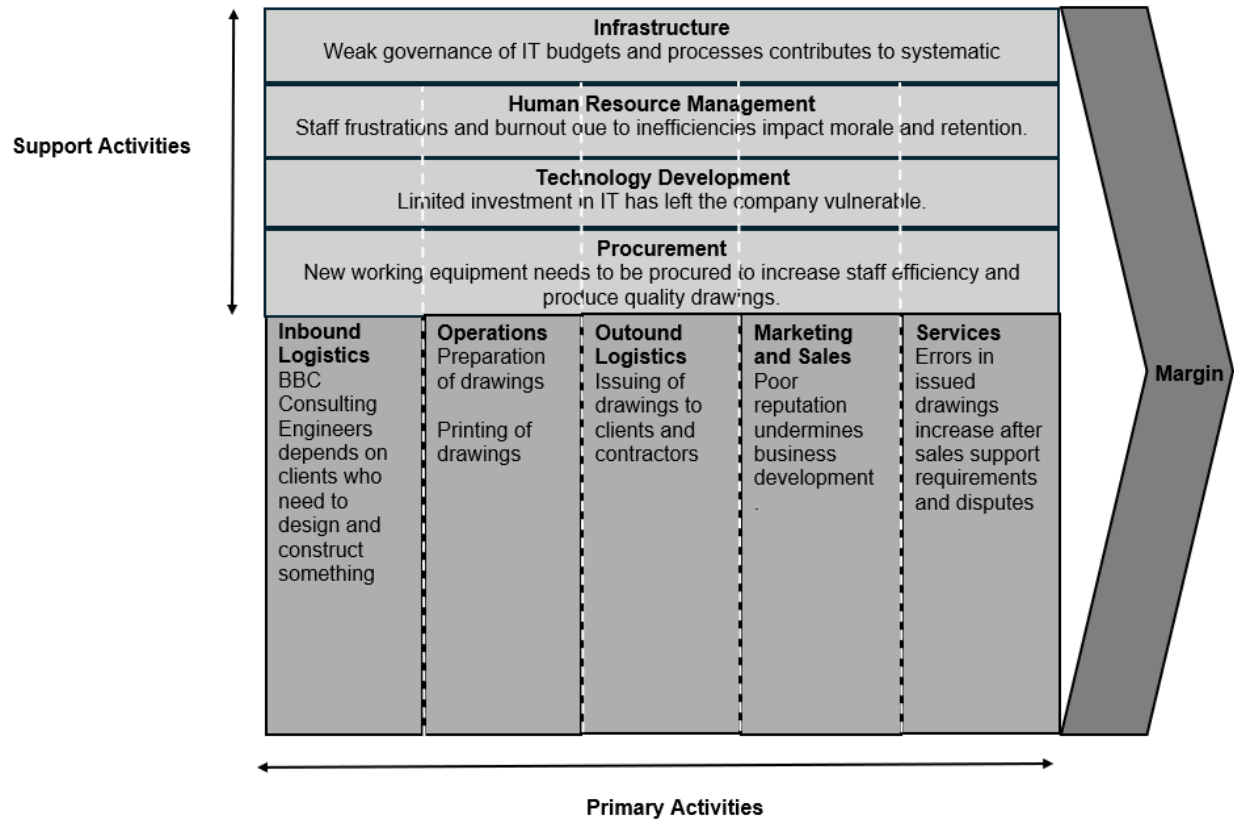
Annexure A (Project Letter)

Annexure B (Business Canvas)

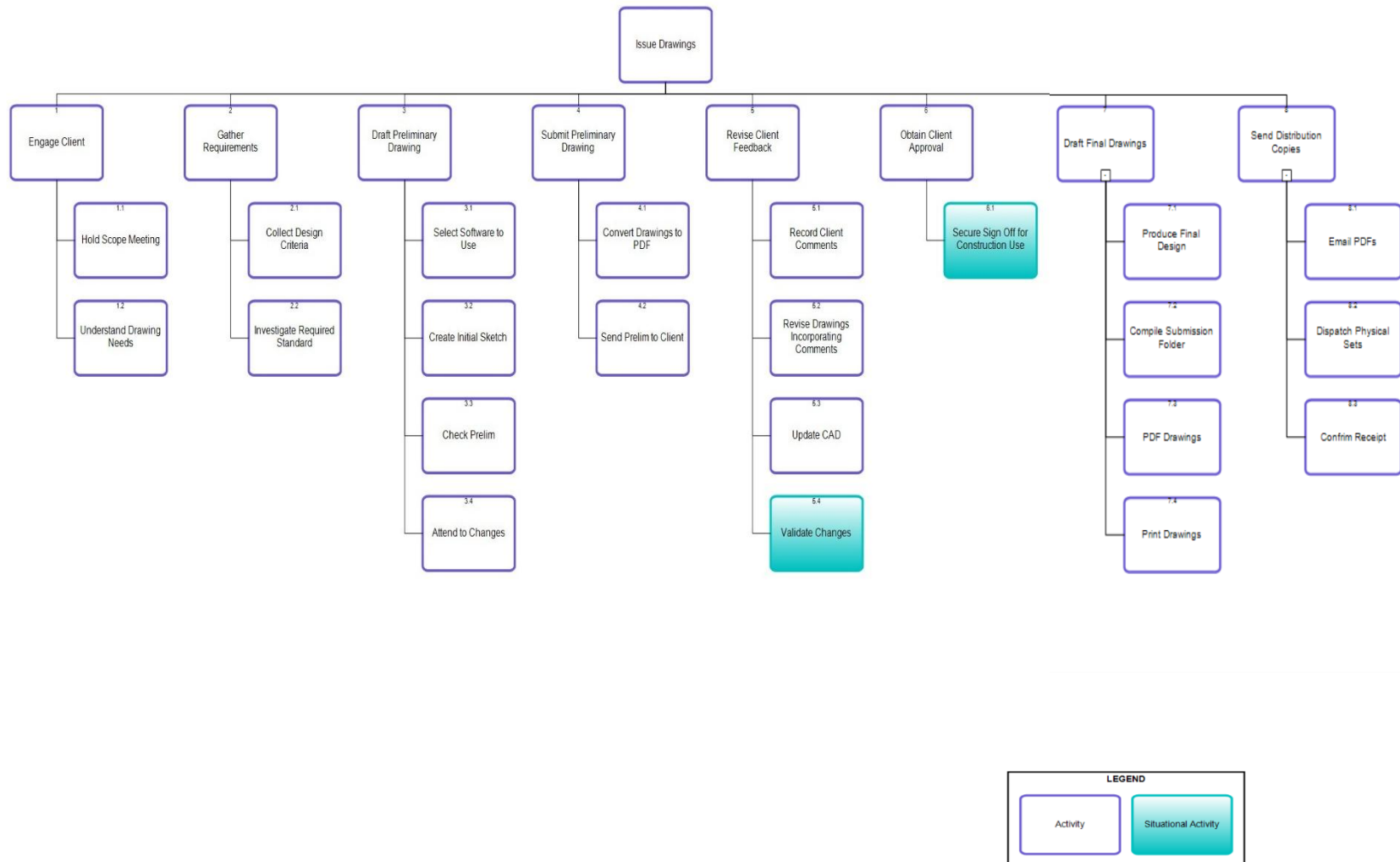
Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
Clients (municipalities, developers), contractors, and IT suppliers. Poor communication and outdated tools weaken partnerships with clients and contractors.	Core activities include drafting, reviewing, approving, printing, and issuing drawings. The inefficiencies in these activities reduce process efficiency and stakeholder satisfaction.	The company's primary value lies in delivering accurate, high-quality engineering designs and drawings that enable clients and contractors to complete infrastructure projects on time and within budget	Healthy and constructive relationship. Understanding that the customer needs to be treated with care and kindness at all times.	Primary customers are municipalities, private developers, and contractors who expect professional service delivery. Dissatisfaction here directly threatens revenue.
	Key Resources		Channels	
	Current resources include AutoCAD 2010, outdated computers, and printers. These resources are not fit for purpose in supporting modern engineering requirements.		Email, courier, and physical meetings. These channels are inefficient for large file transfers compared to modern cloud-based systems.	
Cost Structure			Revenue Streams	

High hidden costs due to rework, wasted time, and equipment maintenance. Investment in updated systems could reduce these costs long-term.	Fee-for-service contracts based on project delivery. Delays in drawings reduce billable efficiency.
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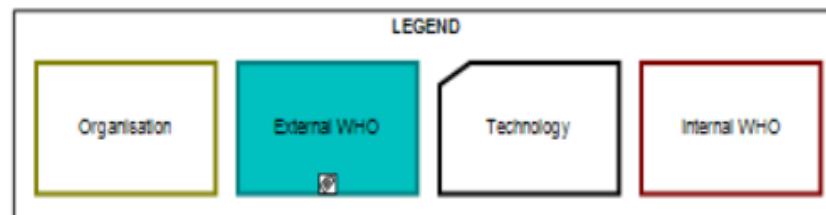
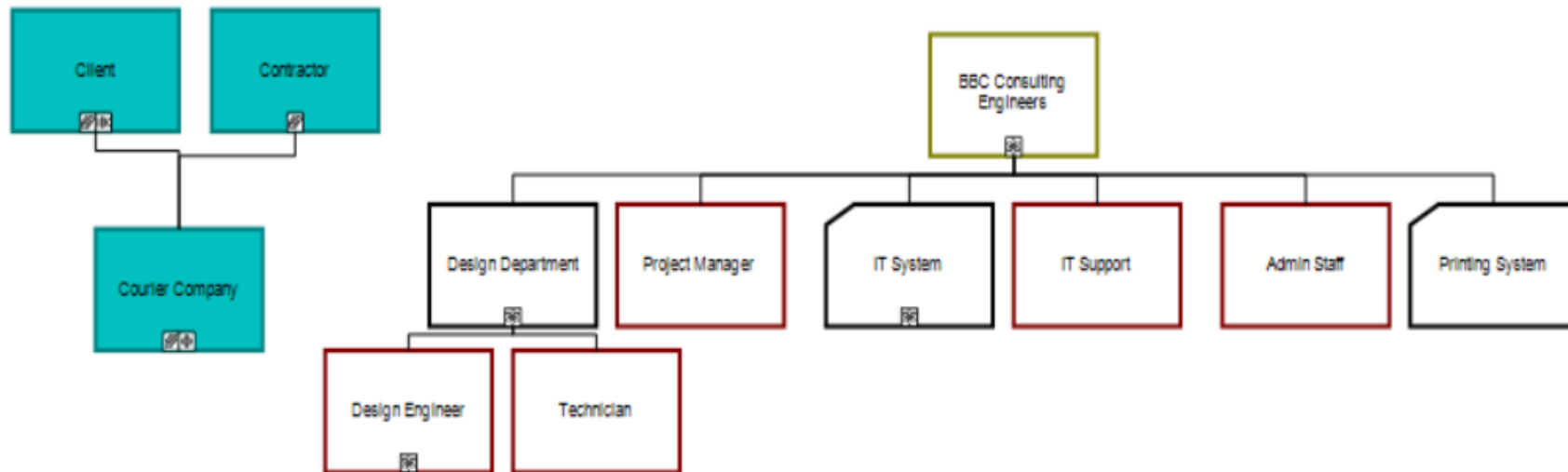
Annexure C (Porters Value Chain)



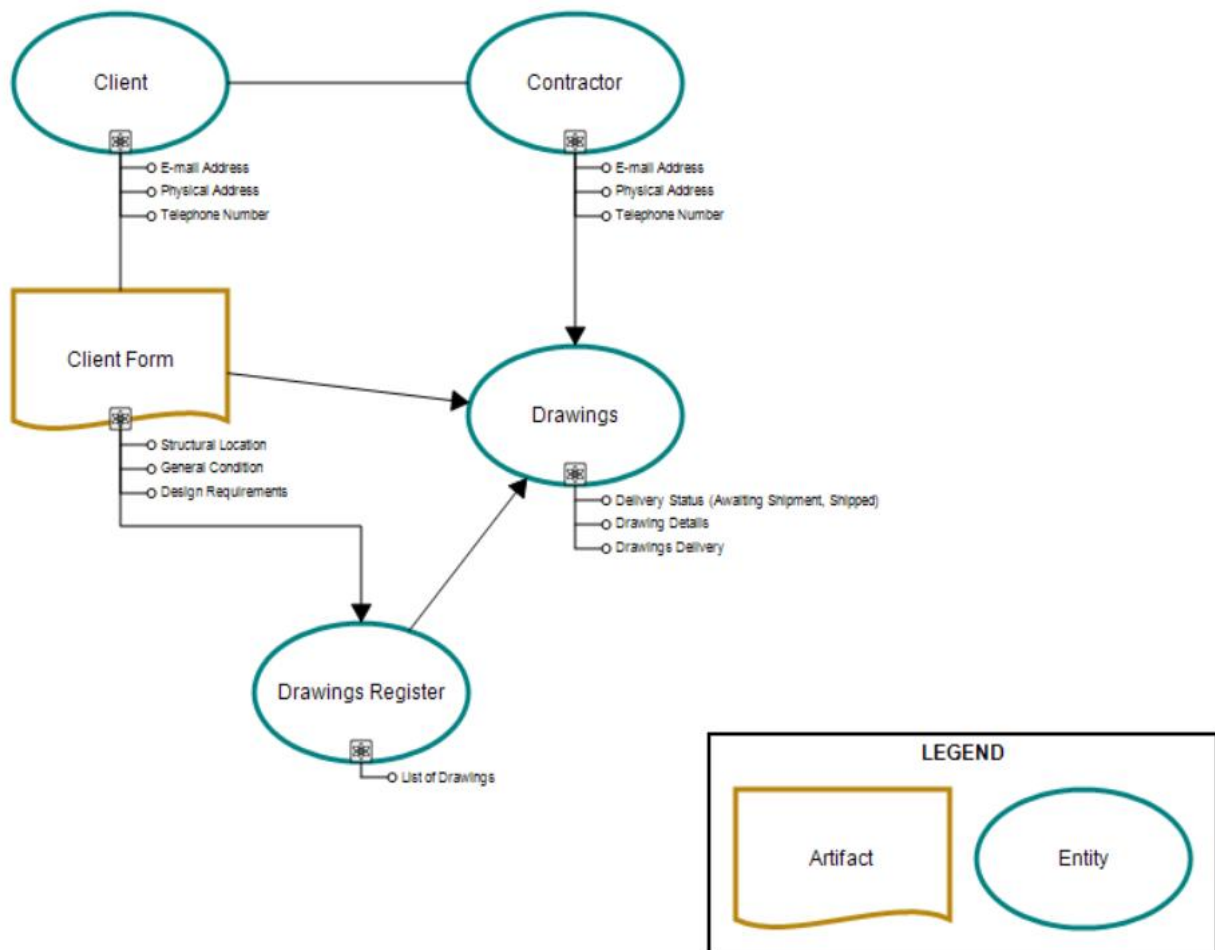
Annexure D (WHAT Model)



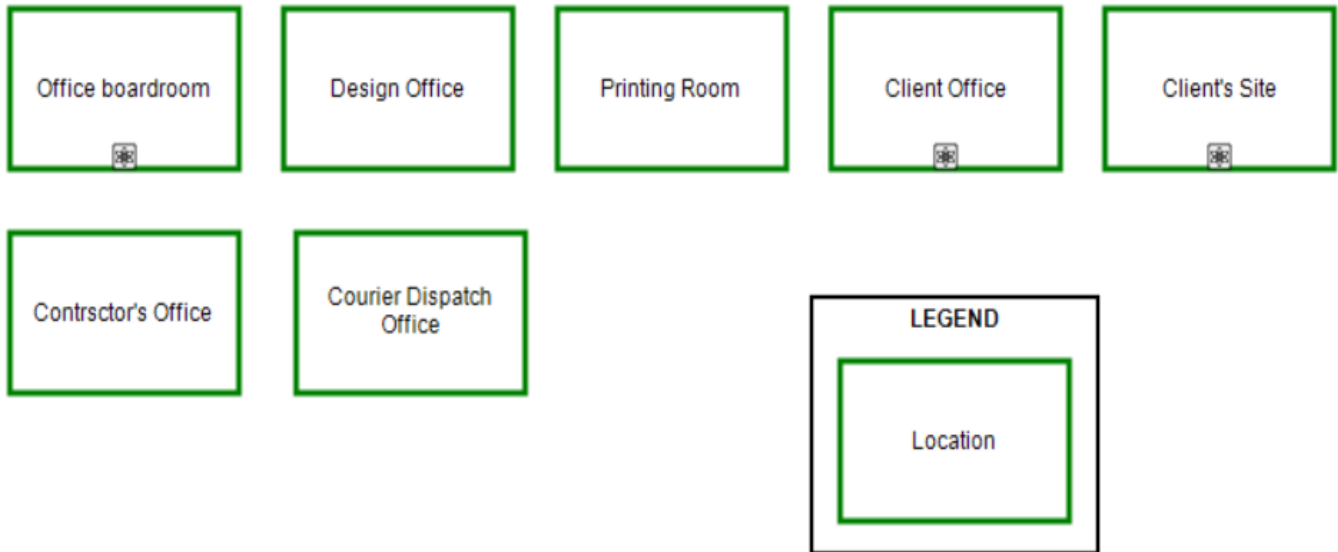
Annexure E (WHO Model)



Annexure F (WHICH Model)

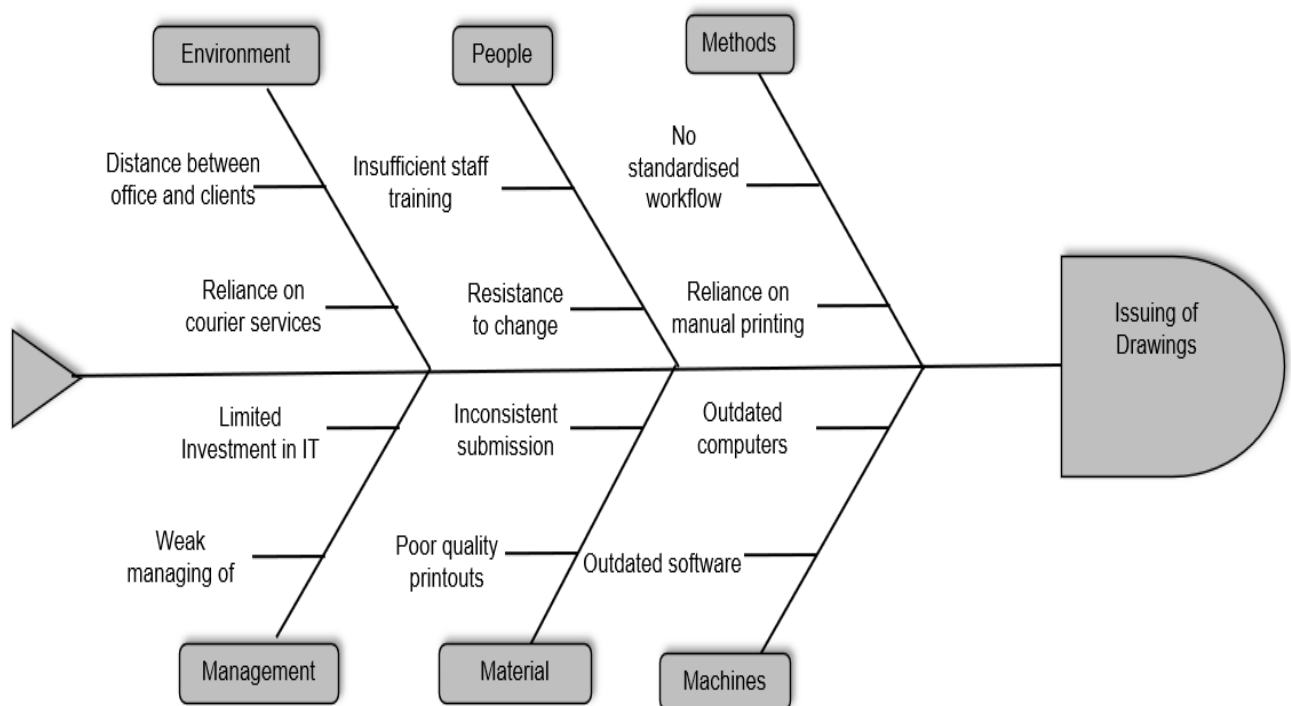


Annexure G (WHERE Model)

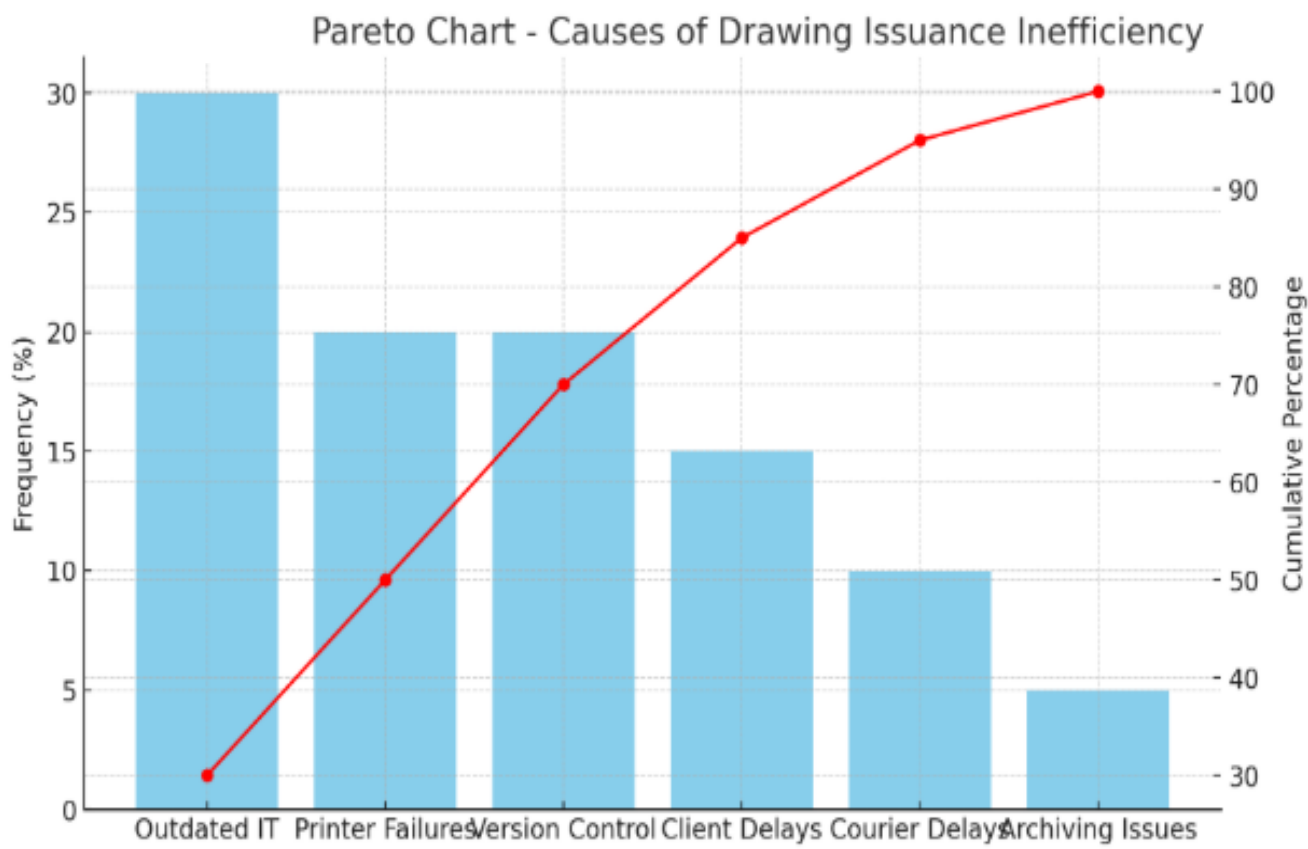


Annexure H (How Model)

Annexure I (Fishbone Diagram)



Annexure J (Pareto Chart)



Annexure K (Mock Interview)

Project: Business Process Analysis: Issuing of Construction Drawings
Organisation: BBC Consulting Engineers
Interview Type: Semi-Structured
Purpose: To capture staff perspectives on challenges, inefficiencies, and opportunities for improvement in the drawing issuance process.

Section A: Background Information

- Interviewee Name: _____
- Job Title/Role: _____
- Department: _____
- Years of Experience: _____

Section B: General Questions

1. Can you describe your role in the drawing issuance process?
2. How often do you engage in issuing or receiving drawings?
3. Which tools or systems do you rely on most frequently?

Section C: Process-Specific Questions

- Client Engagement and Requirement Gathering**
4. How are client requirements typically captured?
 5. What challenges do you encounter when communicating with clients?

- Preliminary Drawing Preparation**
6. What software and hardware do you use when preparing drawings?
 7. Do you experience delays or technical issues at this stage? If yes, please explain.

- Submission and Revision**
8. How are preliminary drawings usually shared with clients?
 9. What challenges arise when incorporating client feedback?

Approval**Process**

10. What is the usual procedure for obtaining client approvals?
11. Are approval timelines predictable or often delayed? Please elaborate.

Final**Production****and****Distribution**

12. How are final drawings produced and distributed to clients and contractors?
13. Do you face any difficulties with printing, scanning, or couriering documents?

Archival

14. How are issued drawings archived?
15. Can you easily retrieve past drawings when needed?

Section D: Improvement-Oriented Questions

16. In your opinion, what are the main bottlenecks in the current process?
17. What tools, systems, or changes would make your work easier?
18. How do you feel about adopting new digital platforms for issuing drawings?
19. What training or support would you need to use new systems effectively?

Section E: Closing

20. Is there anything else you would like to share about your experience with the drawing issuance process?

Annexure L (Log Book)

Hours Log Book		
Name:	Luthando Shezi	
Student no:	25S6517	
Date	Hours	Task Description
14-Aug-25	7	Initial research planning, topic scoping
15-Aug-25	6	Literature review on project management & BPM
18-Aug-25	7	Business context research, background writing
19-Aug-25	7	Draft Section 1 – Context Setting
20-Aug-25	6	Continue Section 1, organisational background
21-Aug-25	7	Business Model Canvas & Value Chain
22-Aug-25	6	Refine Section 1, problem statement
25-Aug-25	7	Literature review – Business Analysis theory
26-Aug-25	7	Literature review – BPM & Lean Six Sigma
27-Aug-25	6	Literature review – IT-Business Alignment
28-Aug-25	7	Draft Section 2 – Theoretical Foundation
29-Aug-25	7	Complete Section 2 draft
01-Sept-25	7	W5 Models – mapping WHAT
02-Sept-25	7	W5 Models – WHO & WHICH
03-Sept-25	7	W5 Models – WHERE & HOW
04-Sept-25	6	Refine AS-IS process model
05-Sept-25	6	Finalise Section 3 draft
08-Sept-25	7	Issues list & Root Cause Analysis
09-Sept-25	7	Fishbone analysis narrative
10-Sept-25	6	Gap analysis & Why analysis
11-Sept-25	7	Pareto prioritisation
12-Sept-25	6	Draft Section 4 analysis techniques
15-Sept-25	7	Mock interview transcripts
16-Sept-25	7	Mock workshop minutes
17-Sept-25	6	Stakeholder context analysis
18-Sept-25	7	Complete Section 4 draft
19-Sept-25	7	Review and edit Sections 1–4
22-Sept-25	7	Draft Section 5 – Results & Conclusions
23-Sept-25	7	Draft Section 5 – Recommendations & Roadmap
24-Sept-25	6	Refine Section 5 and editing
25-Sept-25	7	Draft Annexures A–C
26-Sept-25	7	Draft Annexures D–G
28-Sept-25	6	Final review, proofreading, summary, formatting

